

TUBUAI, French Polynesia

WMO: 919540

Lat: 23.371S

Lon: 149.529W

Elev: 6

StdP: 101.25

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.2	16.4	9.3	7.3	19.2	10.3	7.8	19.4	13.2	21.5	11.8	21.7	1.7	170	0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.2	29.5	25.3	29.1	25.0	28.6	24.8	26.1	28.3	25.8	28.0	25.5	27.7	4.9	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.5	20.7	27.6	25.1	20.2	27.4	24.7	19.8	27.1	81.1	28.4	79.6	28.1	78.1	27.8	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.8	8.7	13.2	30.2	1.1	0.5	12.4	30.6	11.8	30.8	11.2	31.1	10.4	31.4		
(5)			11.9	26.6	0.8	0.6	11.3	27.0	10.8	27.3	10.4	27.7	9.8	28.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.5	25.7	26.2	25.9	24.8	23.3	22.0	21.3	21.1	21.4	22.2	23.5	24.6		
	DBStd	2.32	1.20	1.14	1.25	1.35	1.58	1.56	1.73	1.67	1.58	1.45	1.43	1.28		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	2	0	0	0	0	0	0	1	1	1	0	0	0		
	CDD10.0	4923	486	453	494	446	412	361	351	344	341	378	406	453		
	CDD18.3	1884	227	219	236	196	154	111	94	86	91	119	156	195		
	CDH23.3	10281	1812	1935	2003	1261	608	231	137	111	133	244	665	1142		
	CDH26.7	1147	238	341	339	115	13	1	0	0	1	1	21	77		
(14) Wind	WSAvg	4.5	4.5	4.6	4.2	4.5	3.8	4.5	4.7	4.9	4.8	4.6	4.5	4.6		
(15) Precipitation	PrecAvg	1861	189	200	188	176	147	117	137	137	105	110	133	206		
	PrecMax	2839	535	581	574	447	435	331	299	326	330	297	346	603		
	PrecMin	1191	20	26	22	30	10	8	36	41	10	10	7	14		
	PrecStd	351	103	116	117	95	90	64	69	73	78	65	74	120		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.7	30.2	30.2	29.2	27.7	26.6	26.1	25.9	26.5	26.7	28.1	29.1		
		MCWB	25.2	25.6	25.8	25.1	24.5	23.8	23.2	22.6	23.0	23.2	24.0	25.3		
	2%	DB	29.0	29.6	29.5	28.6	27.0	25.8	25.2	25.1	25.5	25.9	27.2	28.2		
		MCWB	24.9	25.3	25.3	24.8	23.9	22.9	22.4	22.2	22.1	22.5	23.5	24.7		
	5%	DB	28.5	29.1	28.9	27.9	26.4	25.2	24.7	24.5	24.7	25.2	26.7	27.6		
		MCWB	24.6	25.0	25.0	24.4	23.5	22.5	21.9	21.7	21.6	21.9	23.1	24.3		
	10%	DB	28.0	28.6	28.4	27.3	25.8	24.7	24.1	23.9	24.0	24.7	26.1	26.9		
		MCWB	24.4	24.8	24.8	24.1	23.1	22.0	21.6	21.2	21.0	21.4	22.9	23.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.2	26.6	26.6	26.2	25.4	24.5	24.0	23.7	24.0	24.4	25.1	26.1		
		MCDB	28.2	29.0	28.9	28.1	26.9	25.9	25.3	24.9	25.6	25.8	26.8	27.8		
	2%	WB	25.7	26.0	26.1	25.6	24.8	23.8	23.3	23.1	23.1	23.6	24.6	25.5		
		MCDB	27.8	28.5	28.4	27.6	26.3	25.3	24.6	24.5	24.5	25.1	26.2	27.3		
	5%	WB	25.4	25.6	25.7	25.1	24.3	23.3	22.8	22.5	22.5	23.0	24.1	25.0		
		MCDB	27.5	28.1	28.1	27.2	25.9	24.8	24.2	24.1	24.0	24.5	25.9	26.9		
	10%	WB	25.0	25.2	25.3	24.7	23.7	22.6	22.2	21.9	21.7	22.4	23.6	24.5		
		MCDB	27.3	27.8	27.8	26.8	25.5	24.3	23.8	23.5	23.6	24.1	25.5	26.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	4.2	4.4	4.1	4.3	4.2	4.5	4.5	4.3	4.3	4.2	4.0		
		MCDBR	4.7	4.8	4.9	4.7	4.7	4.7	4.6	4.7	4.9	4.7	4.7	4.5		
	5% WB	MCWBR	2.3	2.3	2.4	2.5	2.8	2.9	3.0	3.1	2.9	2.7	2.5	2.4		
		MCWBR	3.9	4.2	4.3	4.1	3.8	3.9	3.9	4.0	4.3	3.9	3.9	3.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.377	0.381	0.372	0.378	0.367	0.358	0.353	0.355	0.369	0.365	0.370	0.375		
		taud	2.625	2.615	2.646	2.625	2.631	2.639	2.639	2.625	2.576	2.581	2.601	2.618		
	Ebn at Noon	Ebn at Noon	967	949	932	883	853	841	859	893	920	953	967	969		
		Edh at Noon	102	101	94	89	82	78	80	87	99	103	104	103		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	6.47	6.07	5.47	4.36	3.64	3.18	3.54	4.18	5.19	5.92	6.41	6.04		
	RadStd	RadStd	0.65	0.64	0.54	0.34	0.30	0.21	0.29	0.31	0.40	0.33	0.53	0.54		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page